

MUHAMMAD HASSAN

AI / LLM Application Engineer

Lahore, Pakistan | hasansheikhh987@gmail.com | 0314-4343059 | github.com/hassan-khalid234 | linkedin.com/in/hassan-khalid-4028842a1

SUMMARY

Final-year Information Technology student building a portfolio around Generative AI application engineering — retrieval-augmented generation, LLM integration, and full-stack AI-powered applications, deployed and live, not just prototyped. Comfortable across the stack, from Python backend logic and databases to LLM APIs and workflow automation. Strong analytical thinker, fast learner, and consistent shipper of independently built, well-documented projects.

TECHNICAL SKILLS

- **Programming:** Python (core & OOP)
- **Generative AI & LLMs:** LLM API integration (Groq), Retrieval-Augmented Generation (RAG), prompt design, embeddings, working knowledge of LangChain-based workflows
- **Machine Learning / NLP:** ML coursework — supervised learning (regression, random forest, KNN, Naive Bayes via scikit-learn), unsupervised learning fundamentals, text classification
- **Backend & APIs:** FastAPI, REST API design, Pydantic, SQLite, database design
- **Vector Databases & Retrieval:** ChromaDB (dense retrieval), BM25 hybrid search, cross-encoder reranking
- **Tools & Practices:** Git/GitHub, Streamlit, n8n workflow automation, unit testing, technical documentation

PROJECTS

RAG Copilot — *Python, ChromaDB, Groq API, Cross-Encoder Reranking, Streamlit*

- Built a framework-free hybrid retrieval-augmented generation system combining dense vector search (ChromaDB) with BM25 keyword search, improving retrieval relevance.
- Implemented cross-encoder reranking to refine retrieved context before LLM generation. Deployed live on Streamlit Community Cloud.

Mood Mirror — *Python, FastAPI, React (Vite), NLP*

- Built and deployed an NLP emotion-detection application mapping detected emotion to a corresponding character persona — FastAPI backend on Render, React frontend on Vercel.
- Combined a 7-class emotion classifier with a zero-shot theme classifier for more accurate character matching.

AI Subscription Intelligence System — *FastAPI, SQLite, Groq LLM API, n8n*

- Built an LLM-powered backend system to track and analyze subscriptions, automating workflows with n8n and resolving real integration issues.

CSV Data Processor — *Python (CLI)*

- Built an object-oriented Python command-line tool with nine filter operators and IQR/z-score outlier detection, verified with unit tests.

EDUCATION

Bachelor of Science in Information Technology

University of Management and Technology (UMT), Lahore | Expected Graduation: 2027

RELEVANT COURSEWORK

Machine Learning (elective), Analysis of Algorithms, Database Administration, Network Security, System & Network Administration, IT Project Management